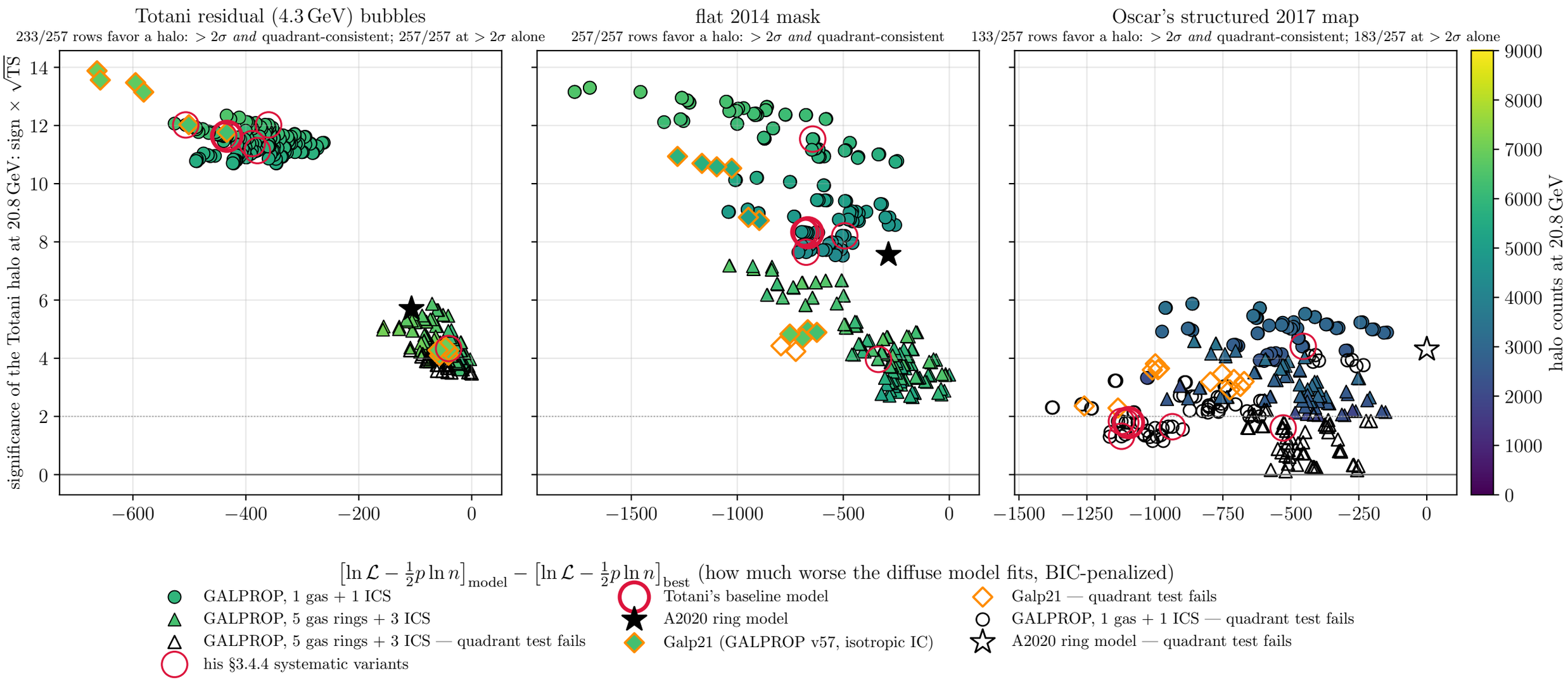

Totani residual (4.3 GeV) bubbles
233/257 rows favor a halo: > 2σ and quadrant-consistent; 257/257 at > 2σ alone
flat 2014 mask
257/257 rows favor a halo: > 2σ and quadrant-consistent
Oscar's structured 2017 map
133/257 rows favor a halo: > 2σ and quadrant-consistent; 183/257 at > 2σ alone
significance of the Totani halo at 20.8 GeV: sign × √TS
halo counts at 20.8 GeV
[ln L − ½p ln n]model − [ln L − ½p ln n]best (how much worse the diffuse model fits, BIC-penalized)
GALPROP, 1 gas + 1 ICS
GALPROP, 5 gas rings + 3 ICS
GALPROP, 5 gas rings + 3 ICS — quadrant test fails
his §3.4.4 systematic variants
Totani's baseline model
A2020 ring model
Galp21 (GALPROP v57, isotropic IC)
Galp21 — quadrant test fails
GALPROP, 1 gas + 1 ICS — quadrant test fails
A2020 ring model — quadrant test fails